

Python Data types

(i) Int

```
In [1]: i = 5      # Integer - No. without Decimal  
i
```

Out[1]: 5

```
In [2]: type()
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[2], line 1  
----> 1 type()  
  
TypeError: type() takes 1 or 3 arguments
```

```
In [ ]: type(i)      # Shows the data type of the Variable
```

```
In [3]: print(i)
```

5

```
In [4]: id(i)       # Shows the memory location of the Variable
```

Out[4]: 140723438937336

```
In [5]: age = 54  
age
```

Out[5]: 54

```
In [6]: print(type(age))  
print(id(age))
```

```
<class 'int'>  
140723438938904
```

(ii) Float

```
In [7]: f = 110.23   # Float - No. with Decimal  
f
```

Out[7]: 110.23

```
In [8]: type(f)
```

Out[8]: float

```
In [9]: id(f)
```

Out[9]: 2127399665296

In [10]: `print(f)`

110.23

(iii) Boolean

In [11]: `b = true`

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[11], line 1  
----> 1 b = true  
  
NameError: name 'true' is not defined
```

In [12]: `b = True` *# Boolean - True or False*
 `b`

Out[12]: True

In [13]: `type(b)`

Out[13]: bool

In [14]: `b1 = False`
 `b1`

Out[14]: False

In [15]: `type(b1)`

Out[15]: bool

In [16]: `p = 20`
 `q = 20`

In [17]: `p = 20, q = 20`

```
Cell In[17], line 1  
    p = 20, q = 20  
    ^  
SyntaxError: invalid syntax. Maybe you meant '==' or ':=' instead of '='?
```

In [18]: `p , q = 20, 20` *# Code Optimization & Memory Optimization*

In [19]: `print(p)`
 `print(q)`

20
20

In [20]: `print(id(p))`
 `print(id(q))`

```
140723438937816
140723438937816
```

```
In [21]: p = p + 10
         p
```

```
Out[21]: 30
```

```
In [22]: print(p)
         print(q)
```

```
30
20
```

```
In [23]: print(id(p))
         print(id(q))
```

```
140723438938136
140723438937816
```

```
In [24]: print(b)
         print(b1)
```

```
True
False
```

```
In [25]: b
```

```
Out[25]: True
```

```
In [26]: b1
```

```
Out[26]: False
```

```
In [27]: True
```

```
Out[27]: True
```

```
In [28]: False
```

```
Out[28]: False
```

```
In [29]: True + False      # Value of 'True' = 1 & 'False' = 0
```

```
Out[29]: 1
```

```
In [30]: True * False
```

```
Out[30]: 0
```

```
In [31]: False / True
```

```
Out[31]: 0.0
```

```
In [32]: False // True
```

```
Out[32]: 0
```

(iv) Complex

```
In [33]: c = 10 + 20j          # Complex - Real no. = 10 ,Imaginary no. = 20 and 'j' is th  
c
```

```
Out[33]: (10+20j)
```

```
In [34]: type(c)
```

```
Out[34]: complex
```

```
In [35]: c1 = 20 + 30j  
c1
```

```
Out[35]: (20+30j)
```

```
In [36]: print(c)  
print(c1)
```

```
(10+20j)
```

```
(20+30j)
```

```
In [37]: c
```

```
Out[37]: (10+20j)
```

```
In [38]: c.real
```

```
Out[38]: 10.0
```

```
In [39]: c.imag
```

```
Out[39]: 20.0
```

```
In [40]: c + c1
```

```
Out[40]: (30+50j)
```

(v) String

```
In [41]: s = Welcome          # String - Contains characters within the quotes
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[41], line 1  
----> 1 s = Welcome          # String - Contains characters within the quotes  
NameError: name 'Welcome' is not defined
```

```
In [42]: s = 'Welcome'      # Single quotes  
s
```

```
Out[42]: 'Welcome'
```

```
In [43]: s1 = "Welcome"    # Double quotes
s
```

```
Out[43]: 'Welcome'
```

```
In [44]: s2 = '''Welcome'''    # Triple Single quotes
s2
```

```
Out[44]: 'Welcome'
```

```
In [45]: s3 = 'Welcome everyone
        I am Priya Darshani Sahu'
s3
```

```
Cell In[45], line 1
      s3 = 'Welcome everyone
          ^
SyntaxError: unterminated string literal (detected at line 1)
```

```
In [46]: s4 = "Welcome everyone
        I am Priya Darshani Sahu"
s4
```

```
Cell In[46], line 1
      s4 = "Welcome everyone
          ^
SyntaxError: unterminated string literal (detected at line 1)
```

```
In [47]: s5 = '''Welcome everyone
        I am Priya Darshani Sahu'''
s5
```

```
Out[47]: 'Welcome everyone \n    I am Priya Darshani Sahu'
```

```
In [48]: s6 = """Welcome everyone
        I am Priya Darshani Sahu"""
s6                                     # Triple Double quotes
```

```
Out[48]: 'Welcome everyone          \n    I am Priya Darshani Sahu'
```

```
In [ ]:
```